Particle Swarm Optimisation for Large-Scale Optimisation

J.M. Ward
Computer Science Department
Stellenbosch University
Stellenbosch, South Africa
24913707@sun.ac.za

***Abstract*—Large scale optimisation problems (LSOPs) present a fundamental challenge for traditional optimisation algorithms due to the curse of dimensionality, a phenomenon exponentially expanding the search space and leading to velocity explosion and premature stagnation. In this report a hybrid particle swarm optimisation (HPSO), integrating both Subspace Initialisation and Stochastic Scaling (two techniques used to better scale LSOPs) is introduced and measured under the hypothesis that the algorithm will outperform standard algorithms on complex multi-dimensional landscapes. By combining both Stochastic Scaling and Subspace Initialisation, an increase in performance is theorised comparing the results of the algorithm to implementations on LSOPs using both approaches separably. Experimental evaluation was performed with dimensionalities 100, 500 and 1000 per configuration. Every algorithm ran for 30 independent runs at 5000 iterations per run.**

I. Introduction

Large Scale Optimisation remains a difficult area in optimisation due to the curse of dimensionality. When the dimension for a problem increases, the search landscape grows exponentially. To help mitigate this scalability issue, approaches such as Subspace Initialisation [3] and Stochastic Scaling [2] have been developed to better navigate high-dimensional problems. The goal of this report is to explore an advanced hybrid particle swarm optimisation (HPSO) model that merges Subspace Initialisation and Stochastic Scaling, and to measure the performance of the model against the algorithms the hybrid model is built from.

Four PSO variants were implemented: Canonical, Subspace Initialisation, Stochastic Scaling, and Hybrid. Each algorithm was rigorously benchmarked against five classic mathematical landscapes: Sphere, Elliptic, Rosenbrock, Rastrigin, and Schwefel 1.2. For every benchmark function, 30 independent runs of 5000 iterations were performed at dimensions 100, 500, and 1000 to ensure statistical significance.

The empirical results demonstrate that the proposed Hybrid PSO significantly outperforms both the Canonical baseline and the independent constituent algorithms of the baseline on the majority of the tested landscapes. By effectively mitigating high-dimensional velocity explosions while maintaining search discipline, the HPSO achieves stable convergence. However, a stagnation observed on the Rosenbrock function highlights an algorithmic trade-off, revealing that the very mechanisms used to address steep landscapes are detrimental on flat-valley terrains.

The remainder of this report is structured as follows. Section 2 introduces the background theory and foundational concepts of PSO. Section 3 details the methodology and mathematical formulations of the advanced variants. Section 4 outlines the empirical procedure and benchmark configurations. Section 5 presents the analysis of the results. Finally, Section 6 provides concluding remarks.

II. Background

Global optimisation focuses on finding the best solution, the global maximum or minimum, for a given objective function over a continuous search space. Unlike local search algorithms, where the search stops in the nearest valley or peak, global optimisation aims to navigate complex landscapes to find the single best solution among many potential local optima. A large scale optimisation problem (LSOP) is a global optimisation problem with a high dimensionality, typically in the hundreds or thousands. This phenomenon is the infamous ”Curse of Dimensionality” in LSOPs. When the dimension in optimisation parameters increase, the search space grows exponentially, making it impossible to scour the entire space for the global optimum. The exponential growth makes exhaustive searches infeasible, and causes distance metrics to be drastically skewed.

Because deterministic or exhaustive approaches fail under these extreme conditions, researchers rely on meta-heuristics, like population based evolutionary algorithms, to find ”good enough” solutions in a reasonable timeframe. A prominent approach is particle swarm optimisation (PSO), originally introduced by Kennedy and Eberhart [1]. Inspired by the social foraging behaviour of bird flocks, PSO maintains a population, or ”swarm”, of candidate solutions called particles. As these particles move through the multi-dimensional search space, the trajectories of the particles are guided by three main forces: the current momentum of the particles, the historical personal best position, and the global best position found by the entire swarm.

The algorithm updates the velocity and position of each particle i at iteration t as

$$V_i^{t+1} = wV_i^t + c_1 r_1 (Pbest_i^t - X_i^t) + c_2 r_2 (Gbest^t - X_i^t) \quad (1)$$

and

$$X_i^{t+1} = X_i^t + V_i^{t+1} \tag{2}$$

respectively. In equation 1, w represents the inertia weight of the particle; the inertia weight controls the momentum of the particle. The cognitive and social acceleration coefficients are represented by c_1 and c_2; the coefficients control how strongly the particle is pulled back to its own personal historic best ($Pbest$), or the global best of the swarm ($Gbest$). Finally, r_1 and r_2 are stochastic vectors sampled uniformly from $[0,1]^D$, introducing the stochastic scaling used to encourage diversity and explore the search space.

The general framework of the canonical PSO is outlined in Algorithm 1.

Algorithm 1 Canonical Particle Swarm Optimisation

Initialise swarm with N particles
for each particle $i \in 1..N$ **do**
    Randomly initialise position X_i within search bounds
    Initialise velocity V_i to randomly scaled small magnitudes
    Evaluate objective fitness $f(X_i)$
    Set personal best $Pbest_i = X_i$
end for
Determine global best $Gbest$ from all $Pbest_i$
while termination condition not met **do**
    for each particle $i \in 1..N$ **do**
        Update velocity V_i using Equation 1
        Update position X_i using Equation 2
        enforce boundary constraints on X_i
        Evaluate new objective fitness $f(X_i)$
        if $f(X_i)$ is better than $f(Pbest_i)$ **then**
            Update $Pbest_i = X_i$
        end if
    end for
    Update $Gbest$ if any new $Pbest_i$ is better than current $Gbest$
end while
return $Gbest$

While canonical PSO performs well on low-dimensional spaces, its velocity updates and uniform random initialisation often lead to premature convergence or stagnation when subjected to the sheer volume of Large Scale Optimisation Problems [2]. To address this challenge, modern literature proposes modifying the standard initialisation and velocity mechanisms [2], [3].

III. Methodology

This section details the algorithm architectures to test against the chosen Large Scale Optimisation Problems. The algorithms were implemented in Python, using the cilpy library.

A. Standard Particle Swarm Optimisation

The Canonical PSO algorithm acts as the baseline in this experiment. The baseline follows the standardised formulation [1] and uniform random initialisation pattern defined in Algorithm 1, ensuring a standard benchmark providing a baseline to measure more advanced scaling methods.

B. Stochastic Scaling Particle Swarm Optimisation

In the Canonical PSO, unique random vectors r_1 and r_2 are applied to every single dimension to encourage diversity and inject random noise. In high-dimensional problems, however, this independent scaling model actually causes the swarm to behave erratically, with a high degree of roaming. To mitigate this, the dimensions for the swarm are randomly partitioned into k sub groups, a technique known as Stochastic Scaling [2].

During every iteration, all dimensions are randomly shuffled into k equal-sized groups, where a single r_1 and r_2 value is then drawn and multiplied with every dimension within its respective group. The value of k is not static; the value starts at one for iteration zero and increases linearly up to the maximum dimensionality D.

C. Subspace Initialisation Particle Swarm Optimisation

Particles are uniformly initialised over the entire search space in the Canonical PSO, in an attempt to acquire a broad understanding of the entire landscape. Unfortunately, in high-dimensional spaces, this initialisation model leads to chaotic velocity explosion, causing particles to overshoot desirable regions and bounce off boundary walls. Subspace Initialisation attempts to counter this effect by modifying how the swarm particles are initially distributed, severely constricting the swarms starting positions to a heavily constrained area. The algorithm calculates the geometric center of the search space, and calculates a one-dimensional line that passes directly through the origin. The swarm is then placed all along this line with the initial velocities of the particles set to zero. By initialising the swarm in this manner the algorithm promotes fine-grained exploitation and mitigates velocity explosion, before the particles begin to fan out and explore wider spaces [3].

D. Hybrid Particle Swarm Optimisation

The Hybrid PSO, as seen in Algorithm 2, combines both aforementioned scaling techniques for LSOPs into a single algorithmic structure. The goal of the hybrid model is to harness the benefits of the structured starting positions of Subspace Initialisation to mitigate velocity explosion at the start, with the dynamic and randomised sub-group velocity updates of Stochastic Scaling [2], [3]. This combination controls particle roaming as well as massive step sizes throughout the experiment.

The swarm is initially seeded entirely using the Subspace approach, providing a structural exploration footing. Then, during the iterative loops, the velocity equations are driven entirely by the linearly scaling k-group dimensionality of the Stochastic model. It was hypothesised that the hybrid structure will outperform both independent algorithms tested individually.

Algorithm 2 Hybrid Particle Swarm Optimisation (Subspace Initialisation + Stochastic Scaling)

Phase 1: Subspace Initialisation
Calculate the geometric center C of the search space bounds
Generate an orthonormal seed vector d via Modified Gram-Schmidt
for each particle $i \in \{1, \dots, N\}$ **do**
  Generate position p_1 along the line $p = t \cdot d + C$ (with margin)
  Generate position p_2 along the line $p = t \cdot d + C$ (strict bounds)
  if $f(p_2) < f(p_1)$ **then**
    Personal best $y_i \leftarrow p_2$, Initial position $x_i \leftarrow p_1$
  else
    Personal best $y_i \leftarrow p_1$, Initial position $x_i \leftarrow p_2$
  end if
  Initial velocity $v_i \leftarrow \mathbf{0}$ {Eliminate initial momentum}
end for
Initialise global best $\hat{y}$

Phase 2: Stochastic Scaling Loop
for iteration $t = 1$ to T_{max} **do**
  Calculate group count k (linearly increasing from 1 to D)
  Randomly partition the D dimensions into k groups
  for each group $g \in \{1, \dots, k\}$ **do**
    Draw shared group scalars $r_{1g} \sim U(0,1)$ and $r_{2g} \sim U(0,1)$
  end for
  for each particle $i \in \{1, \dots, N\}$ **do**
    for each dimension $d \in \{1, \dots, D\}$ **do**
      Let g be the specific group containing dimension d
      $v_{id} \leftarrow w v_{id} + c_1 r_{1g}(y_{id} - x_{id}) + c_2 r_{2g}(\hat{y}_d - x_{id})$
      $x_{id} \leftarrow x_{id} + v_{id}$
      Clamp x_{id} to search space bounds
    end for
    Evaluate $f(x_i)$ and update personal best y_i
  end for
  Update global best $\hat{y}$ if necessary
end for

IV. Empirical Procedure

To evaluate the algorithms, the experimental setup was designed to stress test the Particle Swarm Optimisation variants against increasingly large search spaces to capture how algorithmic performance degenerates or improves with dimensionality.

A. Benchmark Functions

The capability of the algorithm variants was tested against five classic benchmarking optimisation functions, consisting of both unimodal and multi-modal landscapes. The functions were evaluated across dimensions (D) 100, 500, and 1000, and the specific character domains were as follows:

- **Sphere**: A smooth, unimodal, strongly convex landscape. Search domain: $[-100.0, 100.0]^D$.
- **Elliptic**: A highly ill-conditioned function, representing stretched topologies. Search domain: $[-100.0, 100.0]^D$.
- **Rastrigin**: A highly multi-modal function containing numerous periodic local minima. Search domain: $[-5.12, 5.12]^D$.
- **Rosenbrock**: A non-convex, narrow "valley" topology testing local minima avoidance. Search domain: $[-30.0, 30.0]^D$.
- **Schwefel 1.2**: A deeply non-separable unimodal function. Search domain: $[-100.0, 100.0]^D$.

B. Experimental Parameters

Fairness across all algorithms was strictly maintained. Every algorithm executed for exactly 30 independent runs to ensure robust averages. Each single run was strictly terminated after 5000 iterations. The swarm size was tightly constrained to 30 particles, the inertia weight (w) was set to 0.729844, and both the cognitive (c_1) and social (c_2) acceleration coefficients were defined equally as 1.49618. These specific parameters were selected and are based on the constriction factor of Clerc and Kennedy [1]. The parameters are proven to guarantee convergent trajectory behaviour and prevent systemic velocity explosion. Because standard baseline parameters across every algorithm were used, any performance discrepancies measured are attributed to the underlying algorithm architecture.

C. Evaluation Metrics

To determine the efficiency and accuracy of the underlying architectures, final fitness values across all benchmarking functions were logged for all 30 independent runs. The **Mean Error** and **Standard Deviation** are reported under tabular conditions in Table I.

Additionally, logarithmic convergence profiles plotting the generation sequence against the global best fitness evaluations are provided to visually represent optimisation speed and trajectory stagnation characteristics over time.

V. Research Results

This section presents the empirical findings of the study. First, convergence behaviours and structural biases are analysed. Following this, a specific note on the Rosenbrock anomaly is discussed.

A. Convergence Behaviours and Algorithmic Structural Bias

The convergence profiles for $D = 1000$ across the benchmark functions are presented in Figures 1 through 5. Across the Sphere, Elliptic, Rastrigin, and Schwefel 1.2 landscapes, a consistent performance hierarchy emerges.

The Canonical PSO consistently exhibited the poorest performance. Driven by independent, per-dimension random initialisation, the Canonical swarm suffers from severe velocity explosion in high-dimensional spaces, causing particles to

TABLE I
MEAN BEST FITNESS AND STANDARD DEVIATION OVER 30 INDEPENDENT RUNS

Function	D	**Canonical PSO**	**Stochastic Scaling**	**Subspace Init**	**Hybrid PSO**
Sphere	100	1.43e+04 ± 8.98e+03	3.33e+02 ± 1.83e+03	7.55e-08 ± 2.93e-07	**2.73e-11 ± 1.44e-10**
	500	2.82e+05 ± 3.86e+04	2.23e+04 ± 7.48e+03	1.81e+01 ± 3.19e+01	**4.40e-01 ± 7.48e-01**
	1000	1.02e+06 ± 6.98e+04	1.56e+05 ± 3.11e+04	5.34e+01 ± 9.31e+01	**4.98e+00 ± 6.43e+00**
Elliptic	100	8.59e+08 ± 5.15e+08	1.99e+07 ± 6.11e+07	2.54e-04 ± 9.82e-04	**1.38e-09 ± 4.91e-09**
	500	1.31e+10 ± 3.99e+09	3.78e+08 ± 1.89e+08	2.89e+05 ± 8.77e+05	**4.94e+03 ± 8.96e+03**
	1000	3.29e+10 ± 6.32e+09	2.89e+09 ± 5.75e+08	1.58e+06 ± 2.37e+06	**2.52e+05 ± 4.14e+05**
Rosenbrock	100	2.40e+07 ± 4.28e+07	2.01e+02 ± 5.67e+01	8.89e+01 ± 1.93e+00	**8.73e+01 ± 9.90e-01**
	500	1.23e+09 ± 2.69e+08	7.65e+06 ± 3.06e+06	6.04e+02 ± 2.87e+02	**4.99e+02 ± 1.16e+01**
	1000	4.44e+09 ± 4.63e+08	8.52e+07 ± 3.33e+07	1.92e+03 ± 2.06e+03	**1.03e+03 ± 8.94e+01**
Rastrigin	100	6.21e+02 ± 8.48e+01	1.88e+02 ± 3.67e+01	6.30e-01 ± 2.42e+00	**8.69e-09 ± 2.45e-08**
	500	3.96e+03 ± 1.64e+02	1.58e+03 ± 1.46e+02	1.02e+01 ± 2.36e+01	**3.06e-01 ± 4.75e-01**
	1000	9.49e+03 ± 2.30e+02	5.10e+03 ± 2.66e+02	2.77e+01 ± 3.79e+01	**6.20e+00 ± 1.97e+01**
Schwefel12	100	8.93e+04 ± 4.32e+04	1.02e+04 ± 1.38e+04	3.94e+00 ± 6.35e+00	**1.65e-03 ± 7.03e-03**
	500	1.77e+06 ± 3.79e+05	7.22e+05 ± 2.69e+05	1.77e+03 ± 3.86e+03	**7.61e+01 ± 1.70e+02**
	1000	6.89e+06 ± 9.79e+05	3.07e+06 ± 9.26e+05	7.73e+03 ± 2.47e+04	**2.67e+02 ± 4.08e+02**

rapidly expand the search radius of the particles and waste evaluations on out-of-bounds roaming.

The purely Stochastic Scaling PSO successfully dampens this velocity explosion through dimensional coupling, keeping the swarm stable. However, the stochastic scaling algorithm plateaus early in the search process. Because the initial positions of the stochastic scaling swarm are scattered uniformly at random, the coupled velocity updates restrict the reachable space of the swarm, preventing the swarm from navigating to the global optimum from a distant starting point.

The Subspace Initialisation PSO bypasses the initial velocity explosion by seeding the swarm entirely along a one-dimensional line passing through the center of the bounds. It is important to note the global optima of these standard benchmark functions lie exactly at the origin. Therefore, Subspace Initialisation effectively drops the swarm precisely on the target landscape at Iteration 0, providing a massive, head-start. However, because the algorithm subsequently reverts to standard Canonical PSO velocity updates, the subspace initialisation algorithm eventually succumbs to high-dimensional chaos and struggles to fine-tune the exploitation of the swarm.

The **Hybrid PSO** heavily dominates these four benchmarks, but the superiority of the hybrid model is not solely due to this initialisation advantage. While the hybrid PSO shares the exact same origin-drop advantage as the Subspace algorithm, the Hybrid immediately transitions into Stochastic Scaling for the iterative velocity updates. This mechanism actively controls particle momentum, preventing the swarm from exploding off the initial subspace line into chaotic jitter. This biased starting position with damped velocity mechanics results in a staircase convergence, allowing the swarm to converge significantly deeper into the local minima than pure Subspace Initialisation.

Note on the Rosenbrock Anomaly

While the Hybrid PSO dominates symmetrically centered topologies, the algorithm immediately stagnates at roughly 10^3 on the Rosenbrock benchmark as illustrated in Figure 5. This

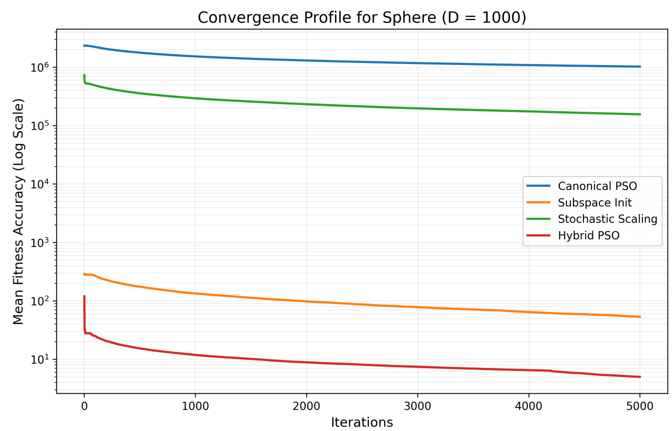

Fig. 1. Convergence Profile for Sphere ($D = 1000$)

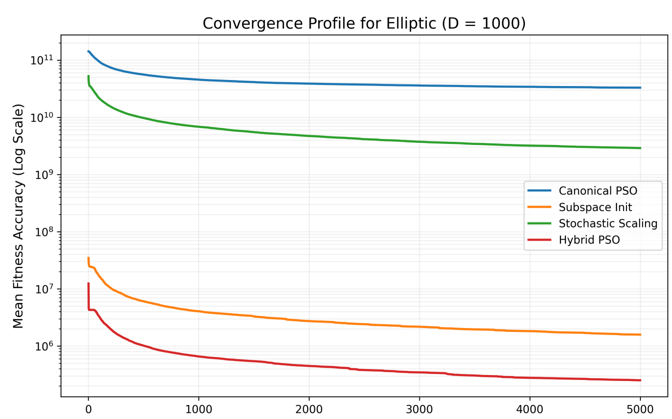

Fig. 2. Convergence Profile for Elliptic ($D = 1000$)

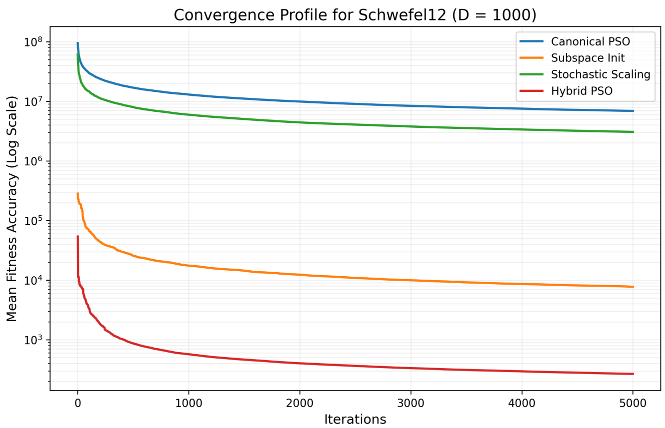

Fig. 3. Convergence Profile for Schwefel 1.2 ($D = 1000$)

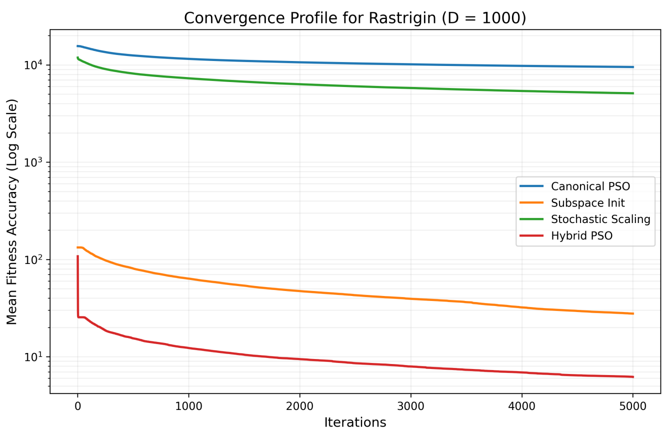

Fig. 4. Convergence Profile for Rastrigin ($D = 1000$)

stagnation occurs due to how the algorithm interacts with the notoriously flat parabolic valley of the Rosenbrock function. Subspace Initialisation forces the swarm to start at the origin $[0, \ldots, 0]$ with zero initial velocity. Trapped on this flat valley floor without initial momentum, and heavily restricted by the dampening of Stochastic Scaling, the particles cannot generate the velocity required to navigate toward the global minimum at $[1, \ldots, 1]$. This anomaly highlights a critical trade-off: mechanisms to prevent velocity explosions on steep topologies actively hinder exploration on flat-valley landscapes.

VI. Conclusion

The primary objective of this study was to evaluate strategies for mitigating the curse of dimensionality in large scale optimisation problems (LSOPs). A hybrid particle swarm optimisation (HPSO) algorithm was developed, integrating Subspace Initialisation and Stochastic Scaling. The performance of this hybrid was benchmarked against Canonical PSO and the independent constituent algorithms of the hybrid across five landscapes at dimensionalities up to $D = 1000$.

The empirical results supported the initial hypothesis for the majority of the tested landscapes. By combining the

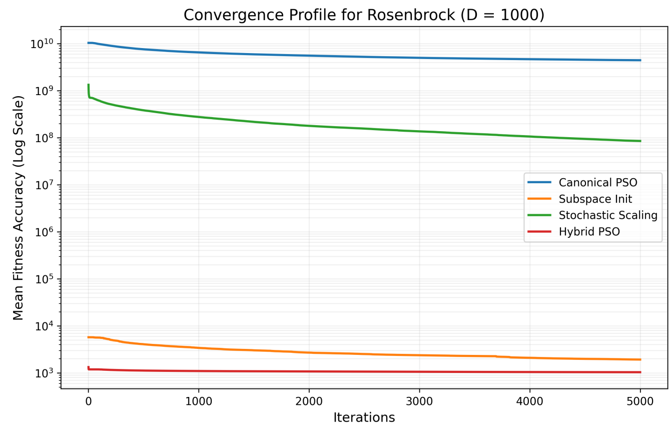

Fig. 5. Convergence Profile for Rosenbrock ($D = 1000$)

geometrically constrained, zero velocity starting conditions of Subspace Initialisation with the coupled velocity dampening of Stochastic Scaling, the Hybrid PSO successfully prevented high-dimensional velocity explosion and excessive particle roaming. The Hybrid PSO outperformed the Canonical, purely Stochastic, and purely Subspace swarms on the Sphere, Elliptic, Rastrigin, and Schwefel 1.2 benchmarks, demonstrating stable convergence.

However, this investigation also identified a critical algorithmic limitation through the Rosenbrock anomaly. The complete stagnation of the Hybrid swarm on the Rosenbrock function proved that while eliminating initial momentum is highly beneficial for symmetrically steep functions, this elimination is actively detrimental on flat-valley landscapes where initial momentum is strictly required for traversal.

Conducting this research highlighted that algorithmic performance in LSOPs cannot be evaluated purely on final fitness metrics. An understanding of how the mechanics of an algorithm interact with the specific spatial landscape of a problem is also important.

Future work should address the initialisation bias present in this study by benchmarking the Hybrid algorithm against ”shifted” landscape variants, where the global optimum is intentionally displaced from the geometric center of the search space. Future studies should investigate adaptive initialisation strategies to detect flat-valley landscapes in order to inject necessary initial momentum to address stagnation vulnerabilities observed in the Rosenbrock landscape.

Appendix A
Acronyms

- HPSO: Hybrid Particle Swarm Optimisation
- LSOP: Large Scale Optimisation Problem
- PSO: Particle Swarm Optimisation

Appendix B
Symbols

- c_1: Cognitive acceleration coefficient
- c_2: Social acceleration coefficient

- C: Geometric center of the search space bounds
- d: Orthonormal seed vector
- D: Dimensionality of the search space
- $f(x)$: Objective fitness function evaluated at point x
- $Gbest$ / $\hat{y}$: Global best position of the swarm
- k: Number of stochastic scaling groups
- N: Total number of particles in the swarm
- $Pbest_i$ / y_i: Personal best position of particle i
- r_1, r_2: Stochastic random vectors drawn from $U(0, 1)$
- t: Current iteration number
- V_i / v_i: Velocity vector of particle i
- w: Inertia weight
- X_i / x_i: Position vector of particle i

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